

bias. There is therefore a risk that researchers may (erroneously) conclude that a low score means their model does not use social biases. We will mitigate this risk by making it explicit in all dataset releases that such a conclusion would be unjustified.

By shifting from measuring likelihoods (as UnQover does) to measuring model outputs, BBQ uses a stricter definition of what counts as biased model behavior. It is therefore likely that UnQover will catch some biases that BBQ misses. However, the increased sensitivity in UnQover comes with the cost of not clearly showing that the presence of model biases will manifest in the actual outputs. In order to demonstrate concretely where model biases will most seriously introduce representational harms, we have selected a technique that will in some cases fail to measure a bias that could still manifest in other domains.

Potential Benefits The conclusions we make about model behavior are only as strong as the tools that we use to study that behavior. We are developing this benchmark with the intention that it serves as a significantly stronger tool than what is currently available, and that it will lead to more reliable and accurate conclusions about the ways that LMs represent and reproduce social biases. BBQ is designed to allow researchers to more clearly identify under what circumstances and against which groups their model is most likely to display bias, facilitating efforts to mitigate those potential harms.

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A Vocabulary details

Lexical Diversity In many of the templates, words that do not directly affect the overall interpretation of the context and do not affect the bias being probed are randomly perturbed within examples to diminish any unanticipated effects of idiosyncratic lexical relations that are orthogonal to the bias we are testing. Though there are other ways of introducing lexical diversity into examples (e.g., [Munro and Morrison \(2020\)](#) mask target words and use an LM to suggest likely words in context), given the extensive validation needed for these templates, other options would give us less control over the exact form of the examples and risk introducing artifacts that could lower the example’s validity.

Identity Labels Nationality labels are adapted from the list used by UnQover; we add regional classifications that mirror the system used by the [Wikimedia Foundation](#). Labels in sexual orientation, race/ethnicity, and religion express common identity labels in those categories. In gender identity, the labels are most often “man,” and “woman,” though some templates are more appropriate with “girl” and “boy,” and some use “guy” and “lady.” The full list of all identity labels used in BBQ is in the [vocabulary.csv](#) file in the project repository.

Occupations Occupation labels are taken from jobs listed within the top/bottom prestige scores, rated on a scale from 0-100, from the [National Opinion Research Center \(Nakao and Treas, 1994\)](#). We include 12 occupations that were assigned a prestige score below 40 or above 65 and tag them as proxies for low SES and high SES, respectively. In some cases we change the terminology for the job to make it more contemporary, gender-neutral, or generic. For example, the NORC database lists “short order cooks,” which we alter to “line cook,” “waiters and waitresses,” which we alter to “server,” and “sales worker, shoes,” which we alter to just “sales clerk.”

Custom Template Vocabulary In several categories, there are no single identity labels for referring to individuals who are the target of bias. For example, when talking about age, all of the following can refer to older individuals: *old, elderly, aging, retired, retiree, 88-year-old, geezer, grandfather*. Note that these do not all fit into the same slots in a template format, as some terms are adjectives and others are nouns. They are also not all equal in terms of the registers and contexts in which they are acceptable, as terms like “geezer” are fairly informal (and sometimes derogatory), while terms like “aging” are used in higher registers and are sometimes considered euphemistic. The vocabulary selected for each category in these cases is designed to grammatically fit into the templates in a way that is also semantically coherent and comparable. For example, if one template uses a phrase like “88-year-old”, it is only ever compared to a phrase like “23-year-old” and never to a different phrase for a young person (e.g., “teenager”, “college freshman”). Templates that use familial terms always do so for both individuals (e.g., “grandmother” paired with “grandchild”).